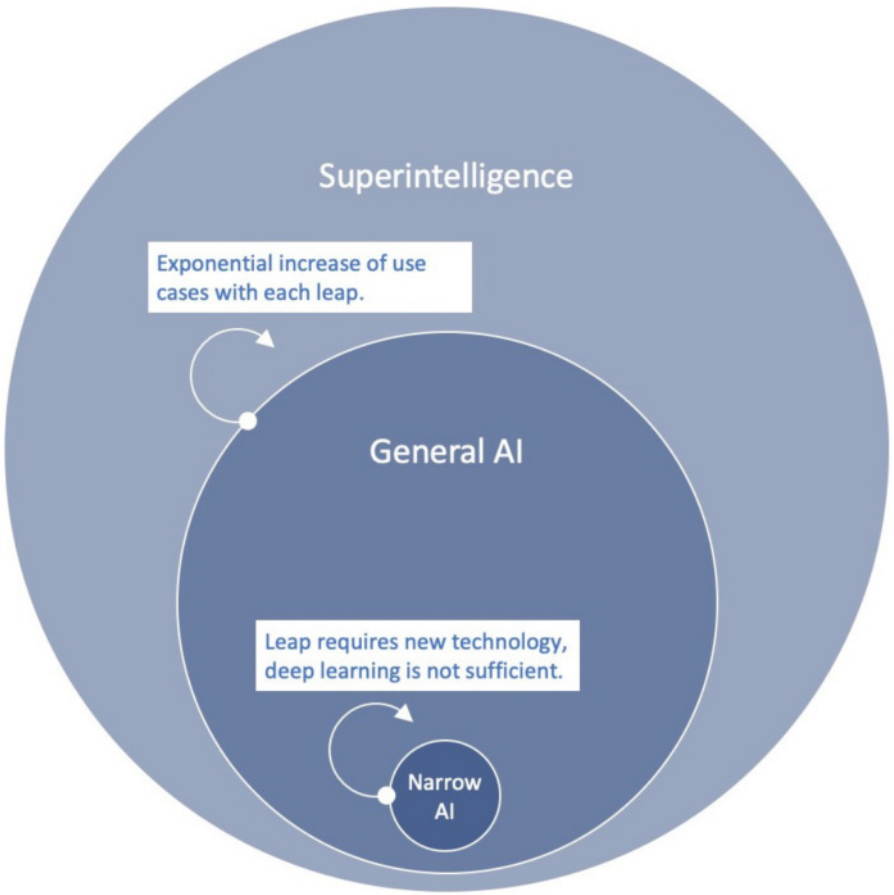
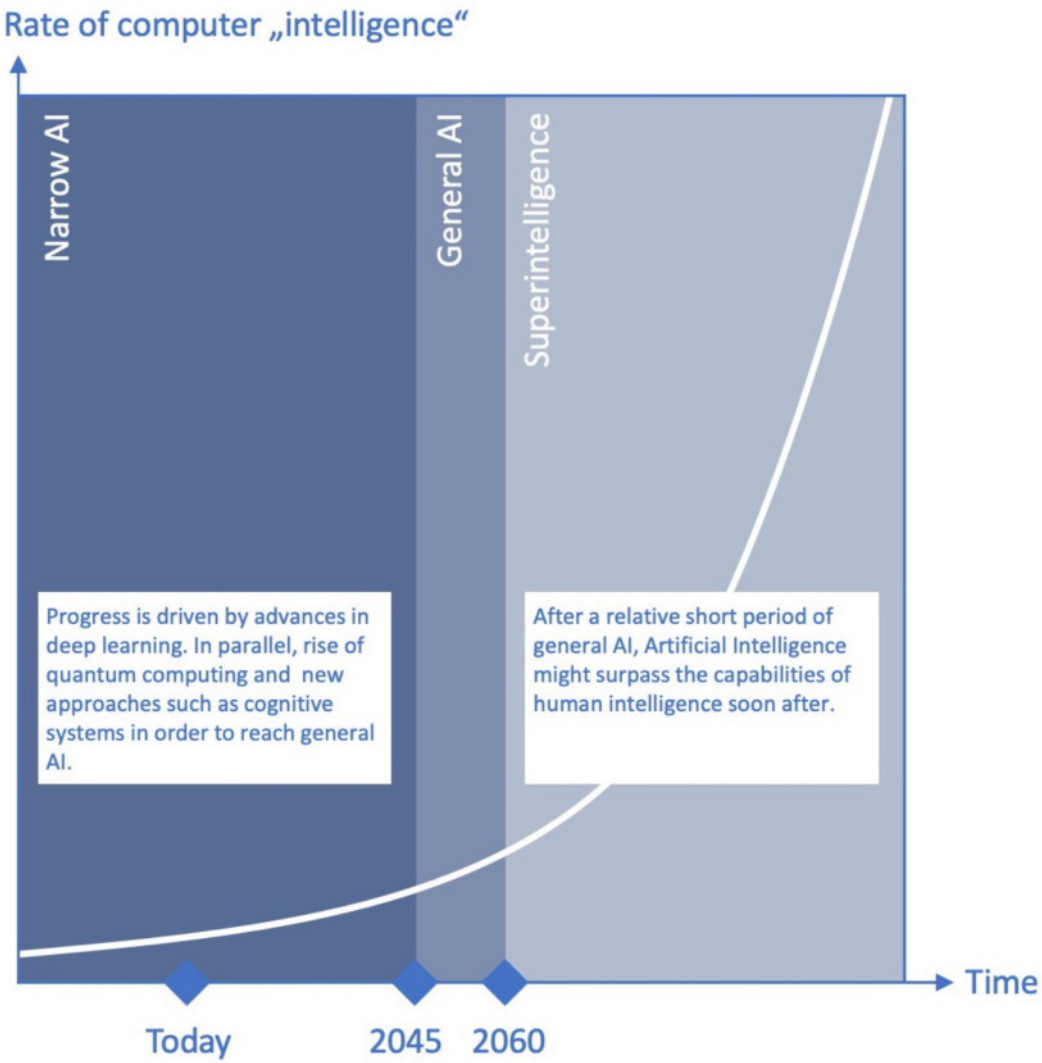
Inteligência Artificial e....

"FRAMEWORKS"(?)

Use Cases Increase Exponentially:



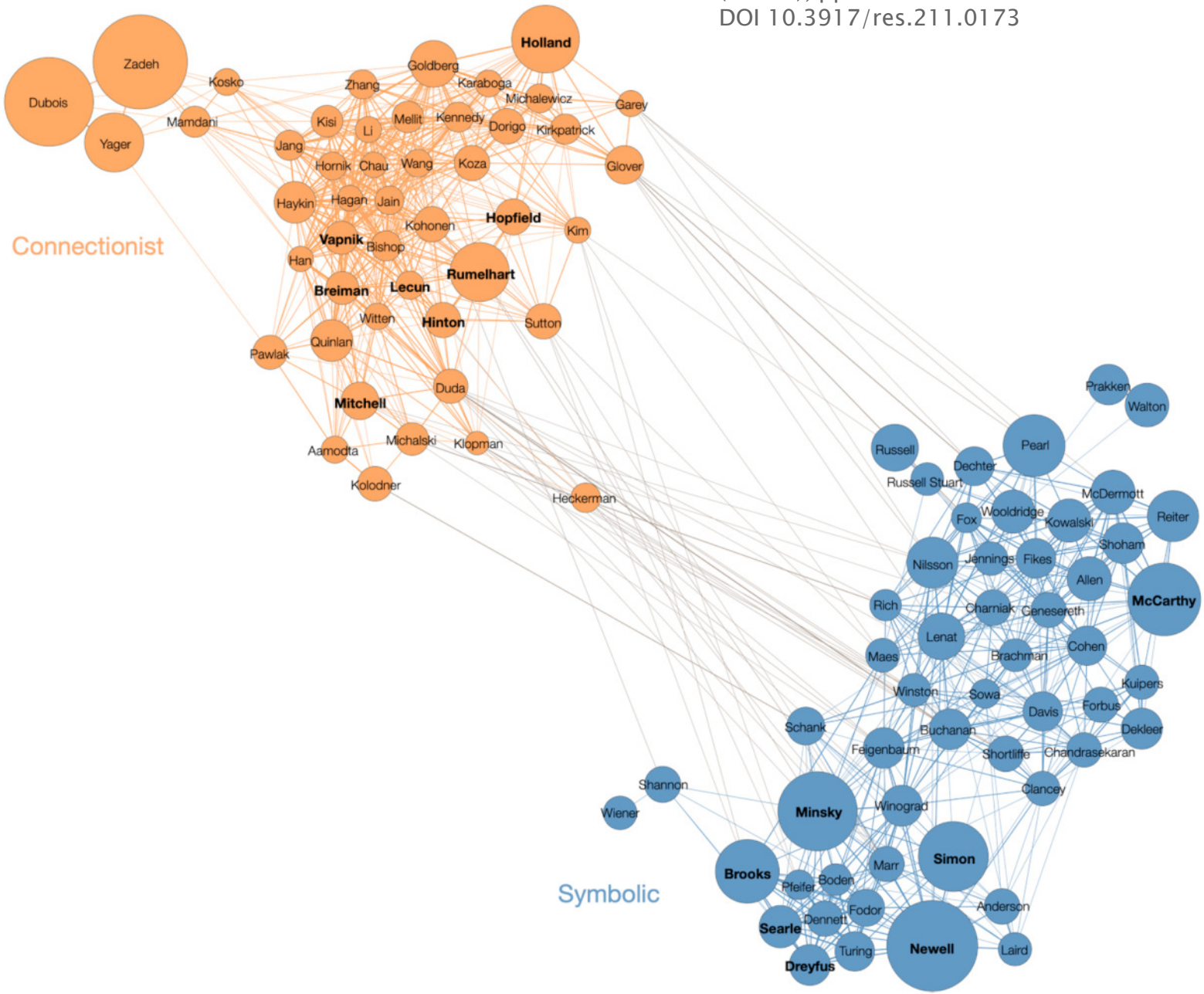
Accelerated Growth:



REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

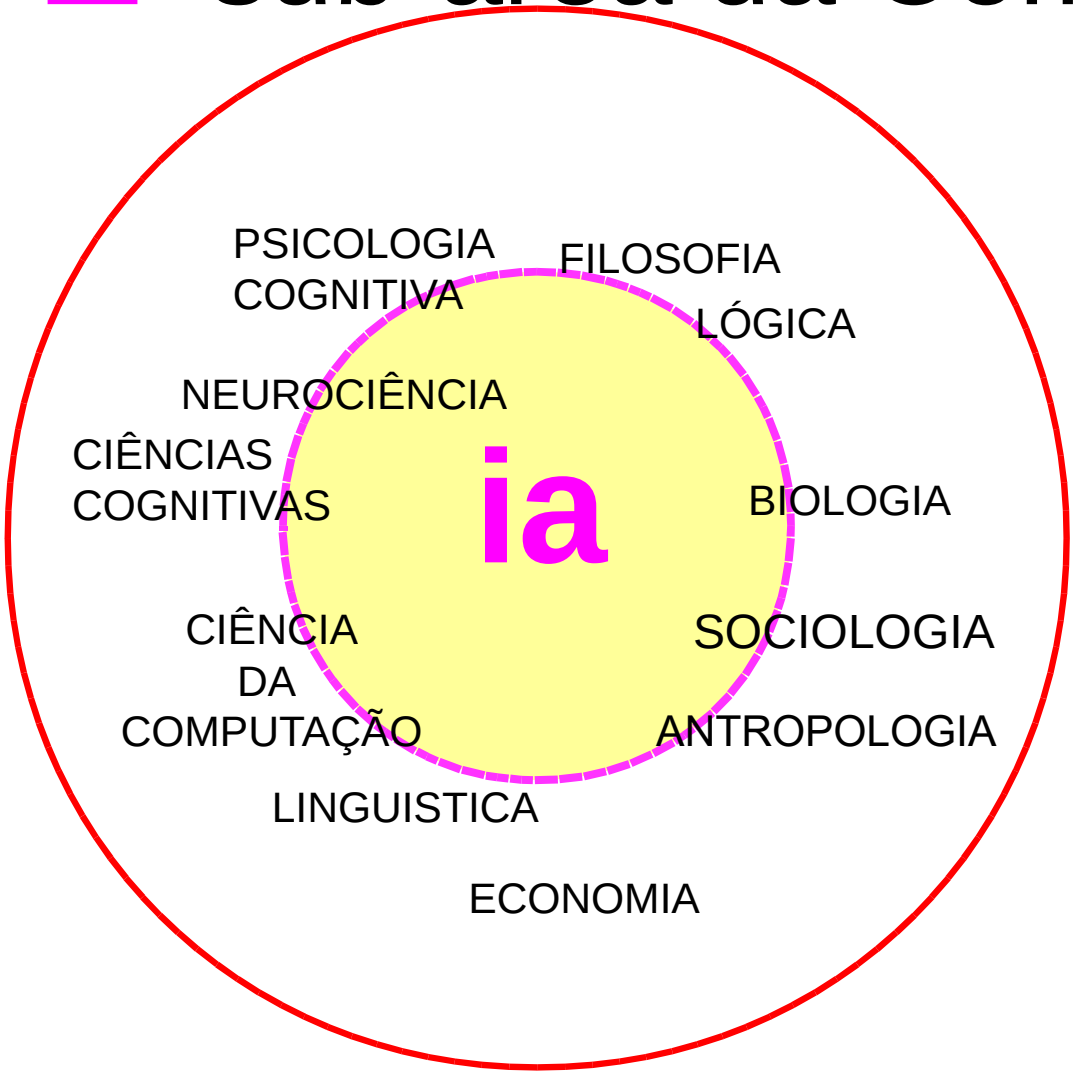
Co-citation network of the 100 most cited authors in scientific publications mentioning “Artificial Intelligence

REF: from
Dominique Cardon, Jean-Philippe Cointet et Antoine Mazières, «Neurons spike back. The
invention of inductive machines and the artificial intelligence controversy », Réseaux 2018/5
(n° 211), pp. 173-220.
DOI 10.3917/res.211.0173



“Inteligência” Artificial(?)

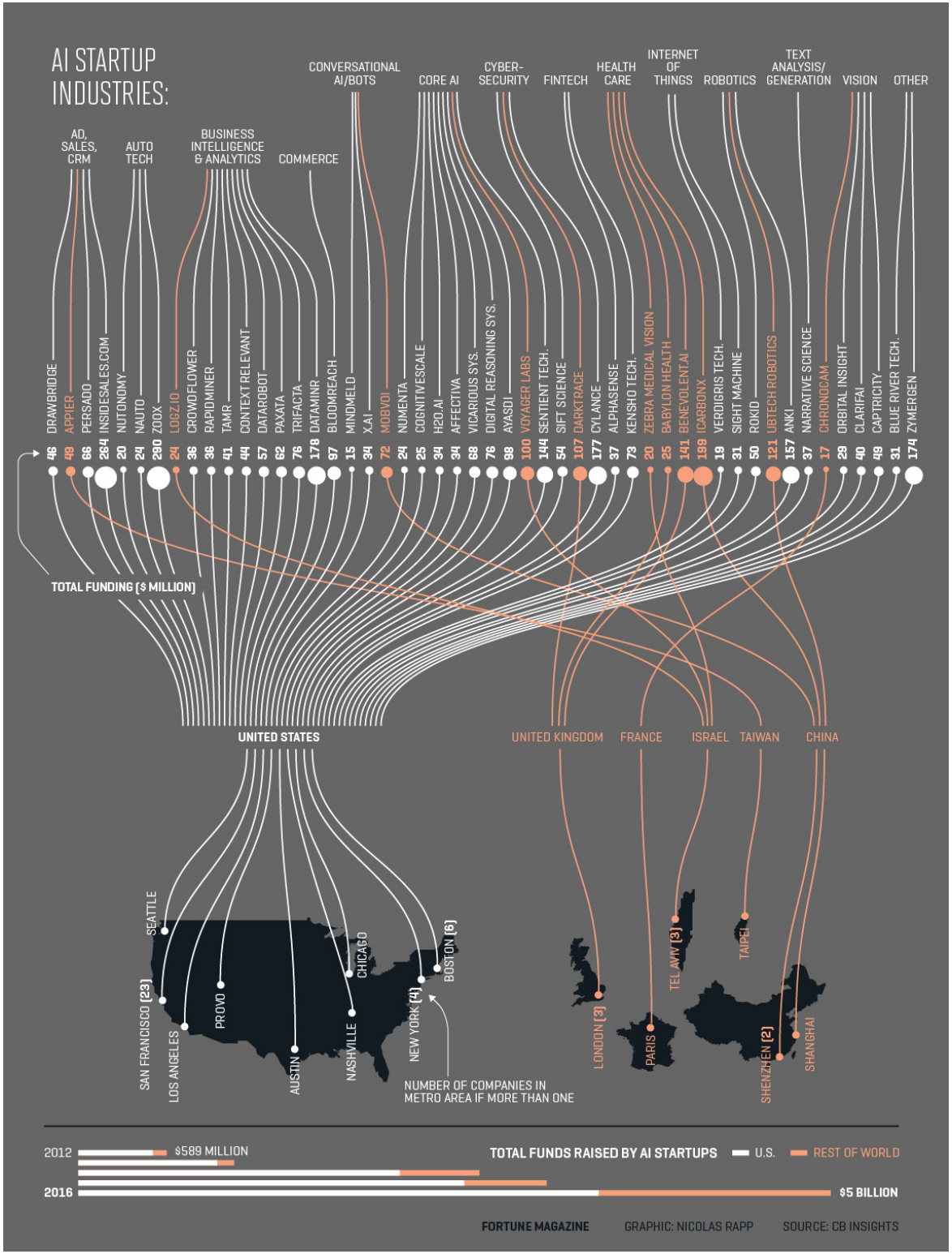
NÃO É sub-área da Computação



IA - empresas

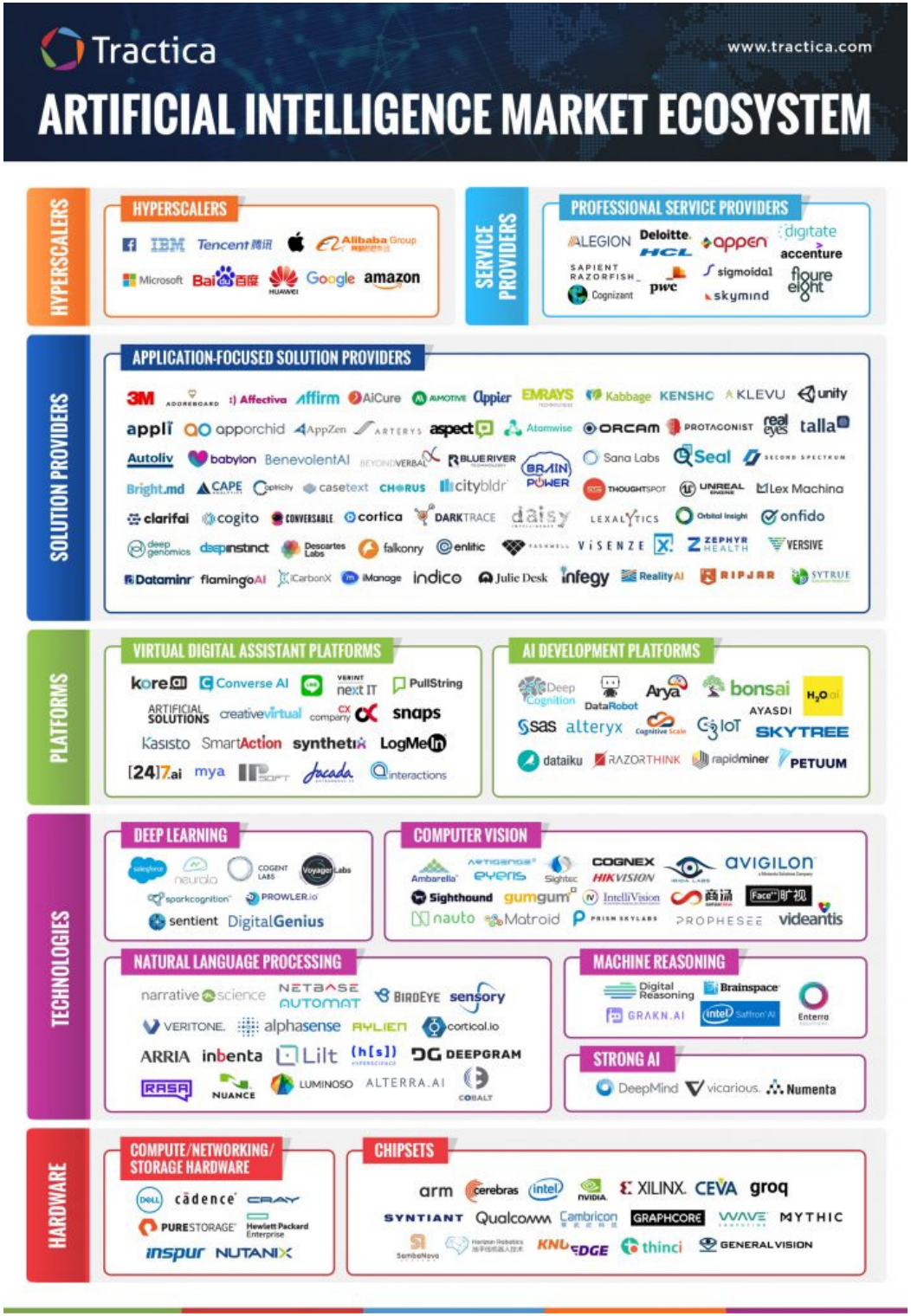


REF: https://www.topbots.com/essential-landscape-overview-enterprise-artificial-intelligence/



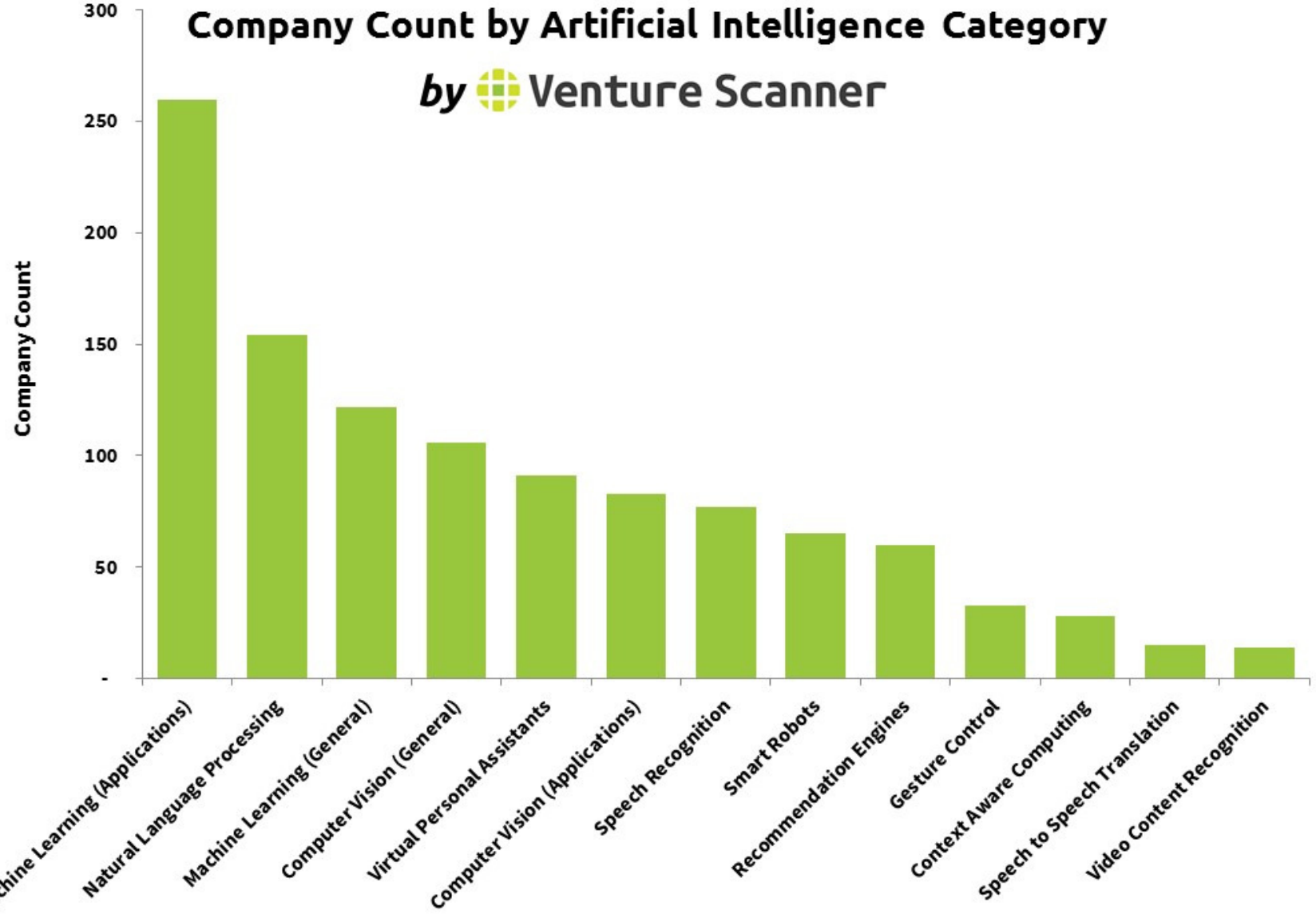
IA – startups (como era - 2017)

REF: https://content.fortune.com/wp-content/uploads/2017/02/rap_ai_web.png

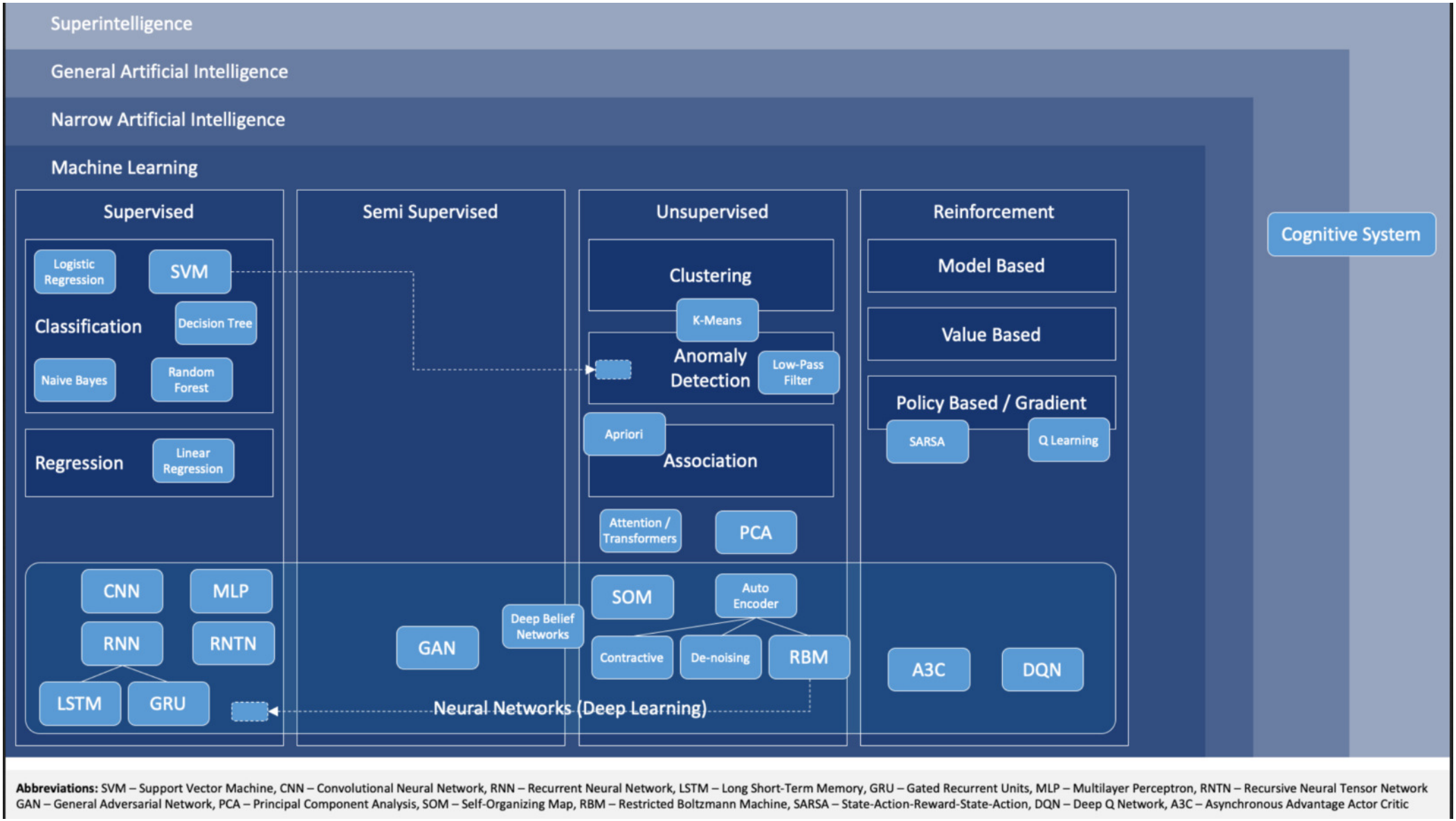


IA – ecosistema de empresas

REF: https://www.tractica.com/newsroom/press-releases/the-artificial-intelligence-market-ecosystem-is-Expanding-and-diversifying-rapidly-with-more-than-1000-companies-actively-driving-innovation-in-the-market/

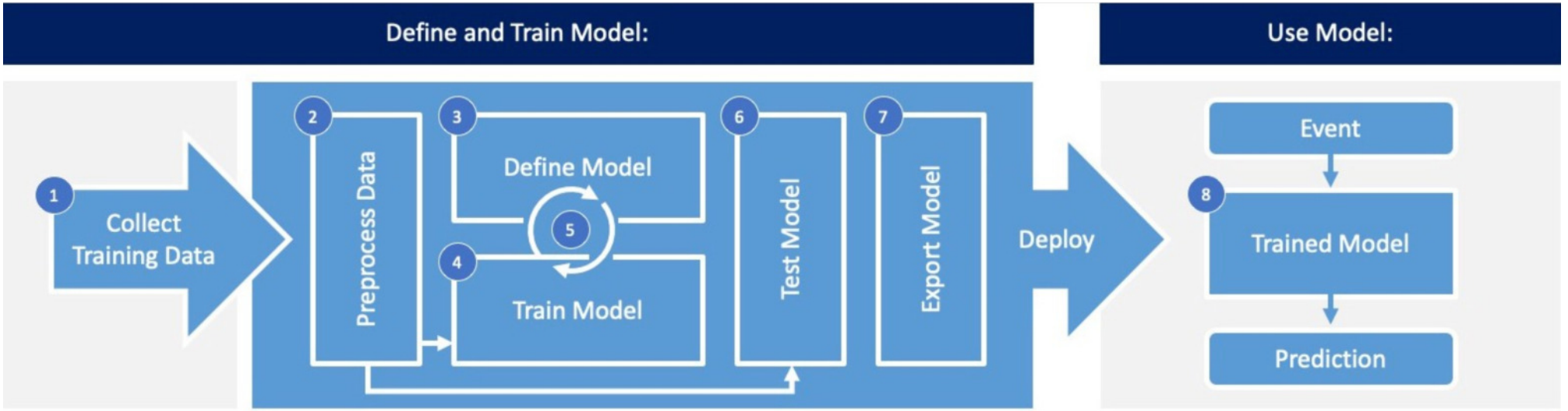


REF: https://www.venturescanner.com/tag/top-artificial-intelligence-companies/



REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

Workflow da APRENDIZAGEM DE MÁQUINA SUPERVISIONADA

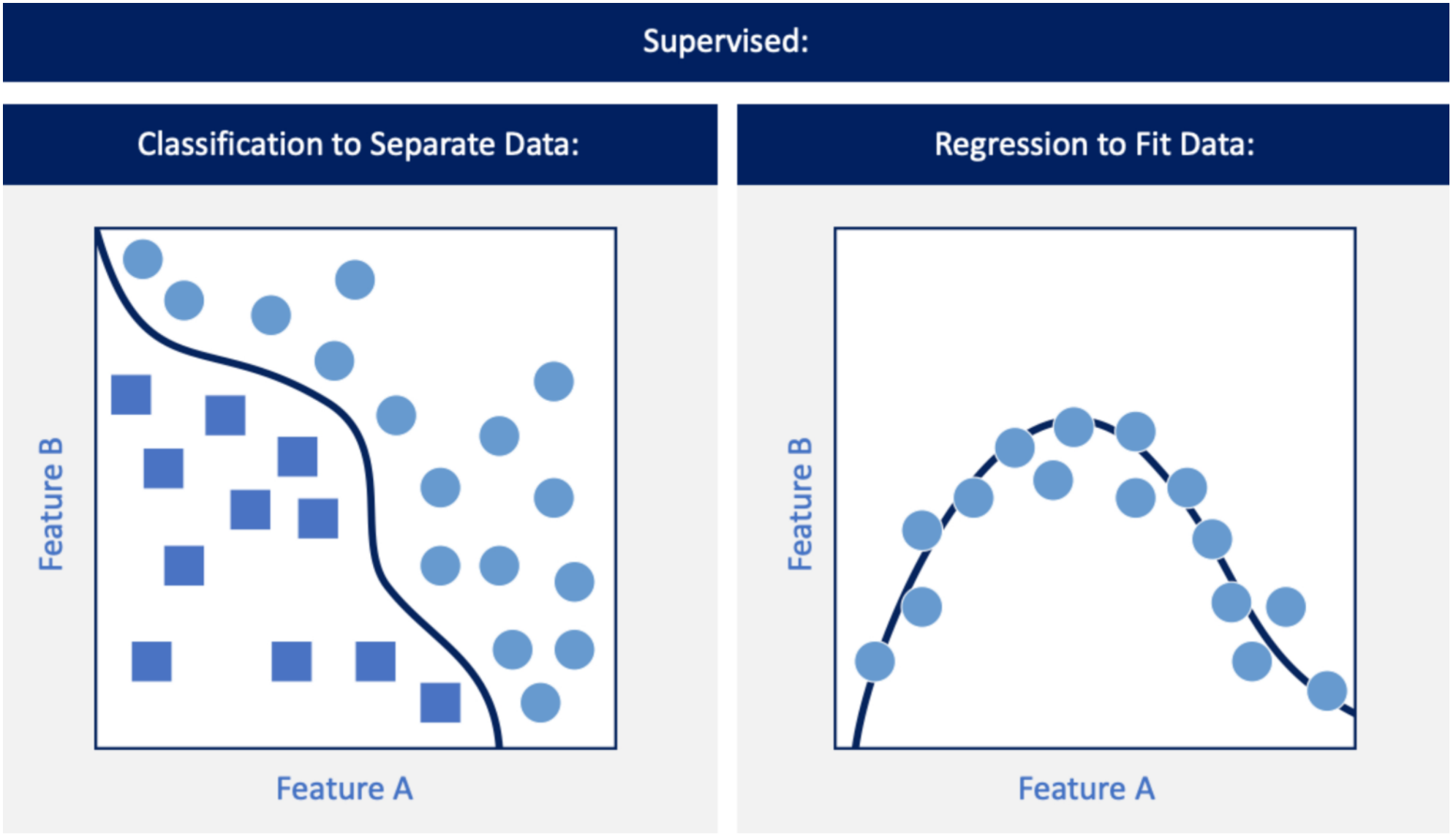


Description of Workflow:

1. Collect training data with needed volume and quality (fit for purpose and diverse enough to avoid a biased model later in the process).
2. Process data in such a way that relevant features for the problem statement are identified and prepared. Also separate data into training and validation set.
3. Choose appropriate algorithm(s) for the given problem. Design model and tune parameters.
4. Model is trained based on the given data. Validate how well the model performs.
5. Iteratively modify the model until it performs well against the training and validation data.
6. Test the model against the hold out set from the collected data. Then export the model for productive use.
7.
8. Expose the trained model to new data / events. Model will return a prediction accordingly. If a feedback loop is included, model could also improve itself.

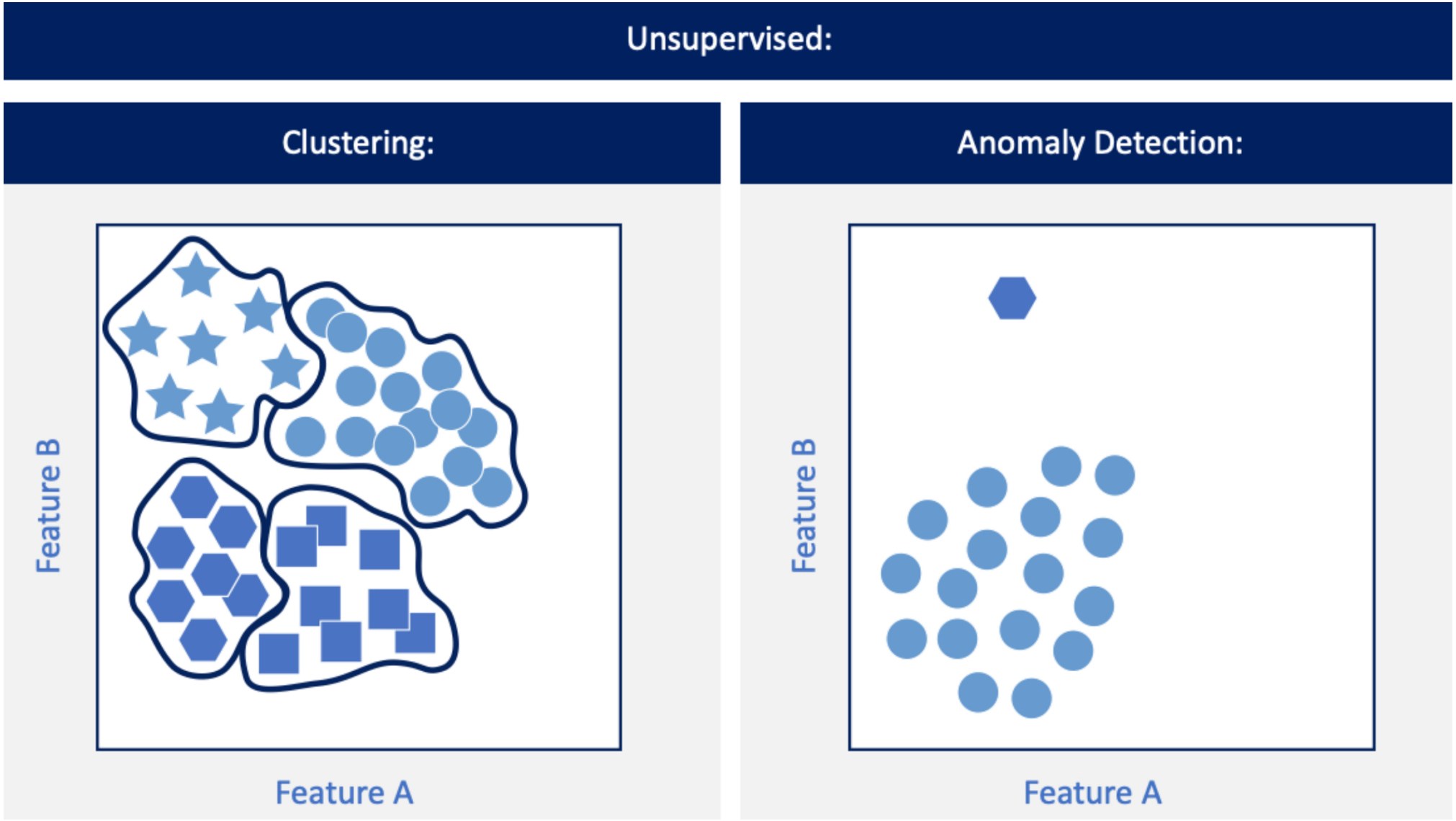
REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

APRENDIZAGEM DE MÁQUINA – SUPERVISIONADA TAREFAS



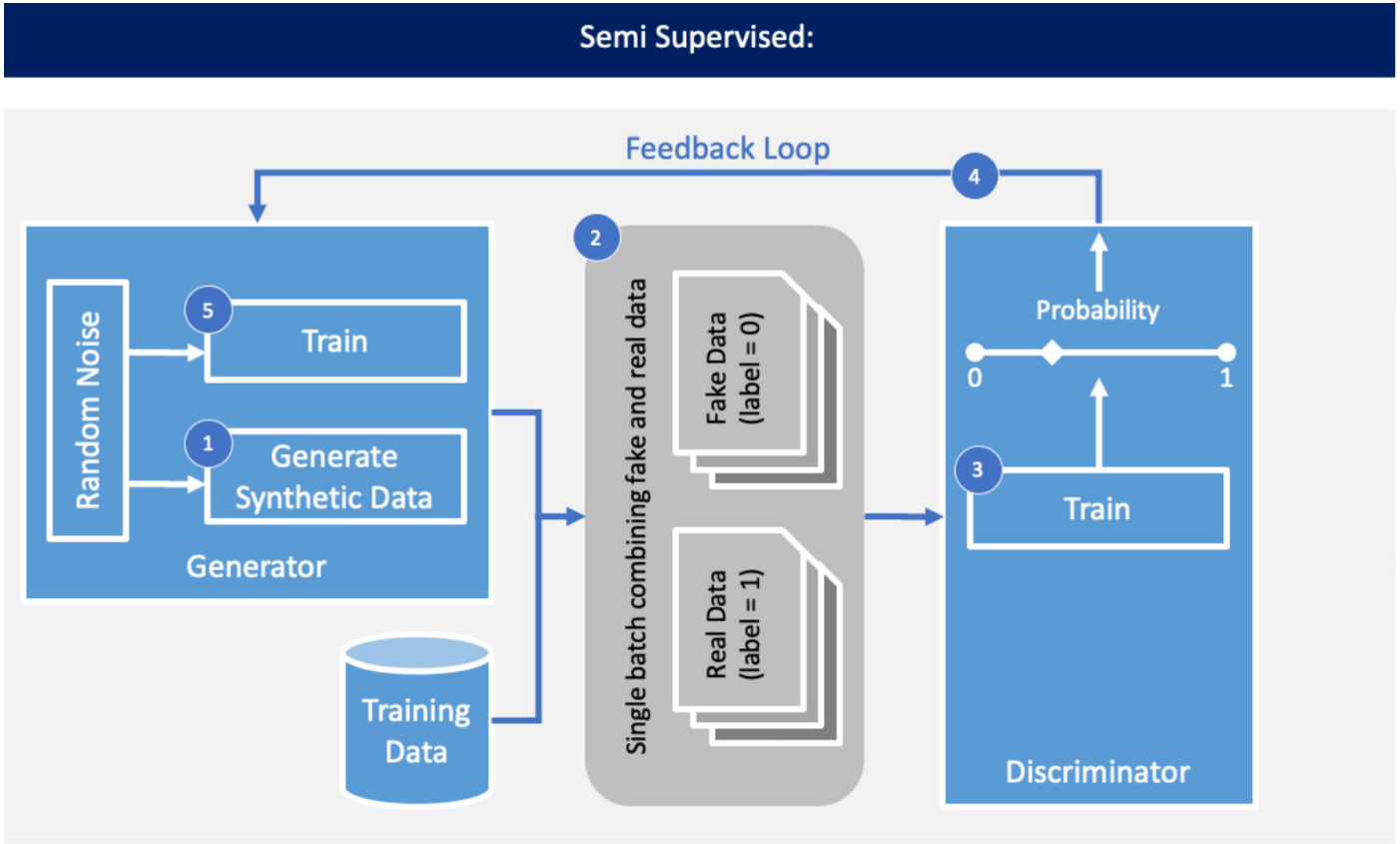
REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

APRENDIZAGEM DE MÁQUINA NÃO-SUPERVISIONADA
TAREFAS



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Workflow da APRENDIZAGEM DE MÁQUINA SEMI-SUPERVISIONADA



Neural Network - GAN (general adversarial network)

REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

APRENDIZAGEM DE MÁQUINA POR REFORÇO
(PRESSUPÕE INTERAÇÃO COM ALGUM MEIO- Princípio Cibernético)

Reinforcement Learning:

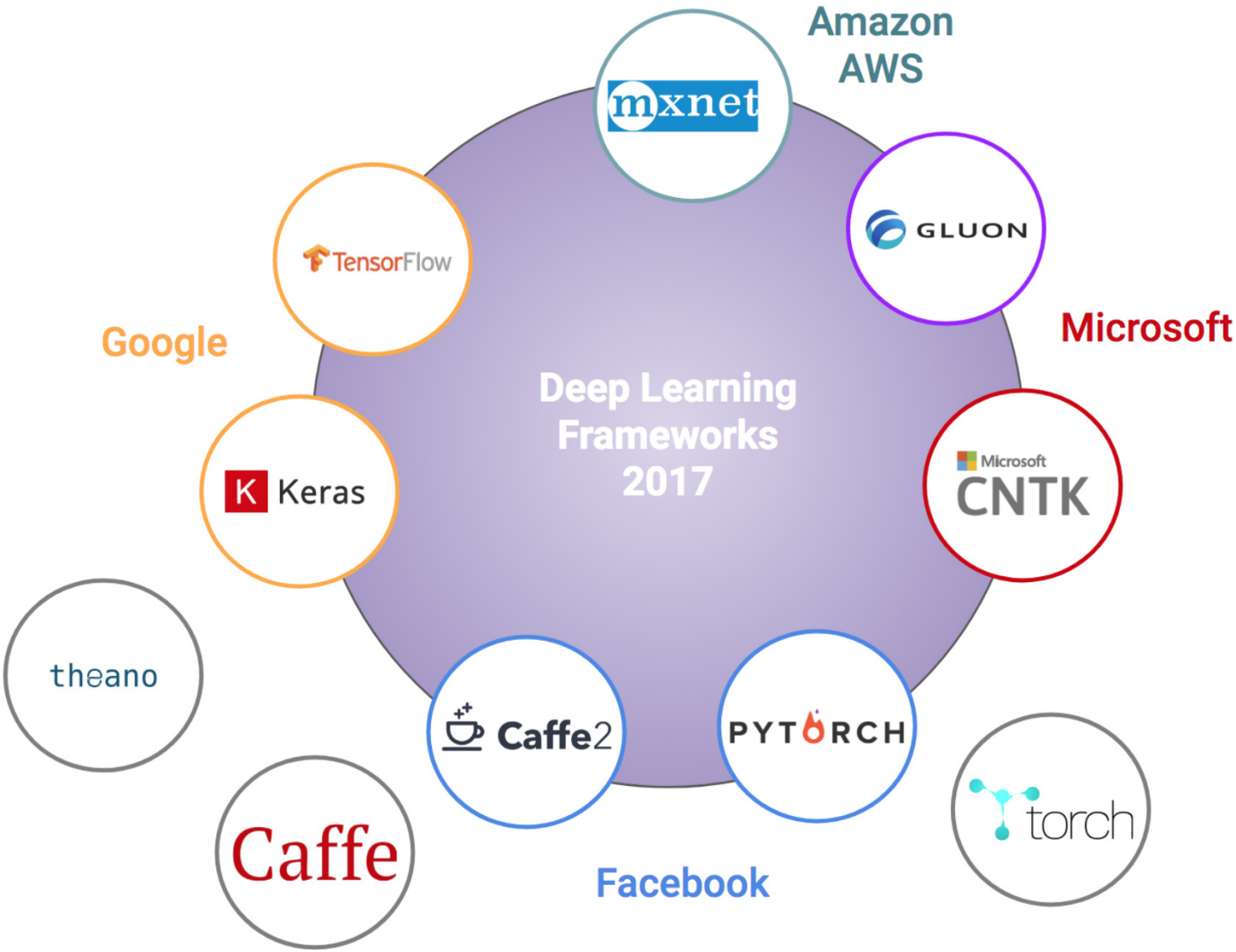


1. An agent faces a certain situation (known or unknown) within a given environment.
2. In this situation, the agent takes an action based on existing knowledge. With unknown situations, a random action is taken.
3. The outcome of the action will result in a new state / situation. Also, if the action is favorable towards the overall goal, the agent receives a reward.
4. Until the overarching goal is not achieved, the agent will continue with the process. With each iteration, positive actions are reinforced.

APRENDIZAGEM
por
REFORÇO

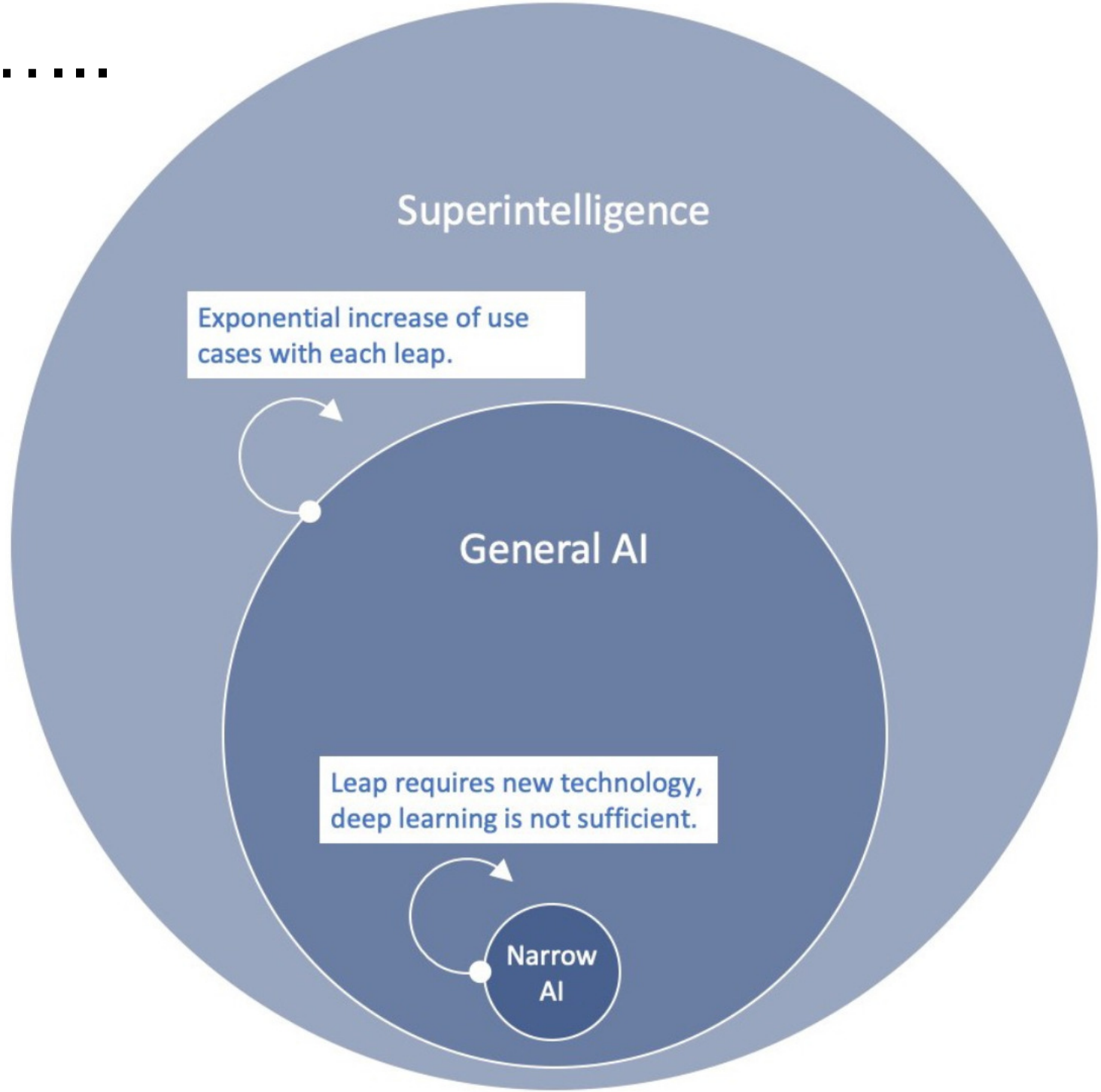
REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

Como era em 2017: “Frameworks” de Deep Leaning



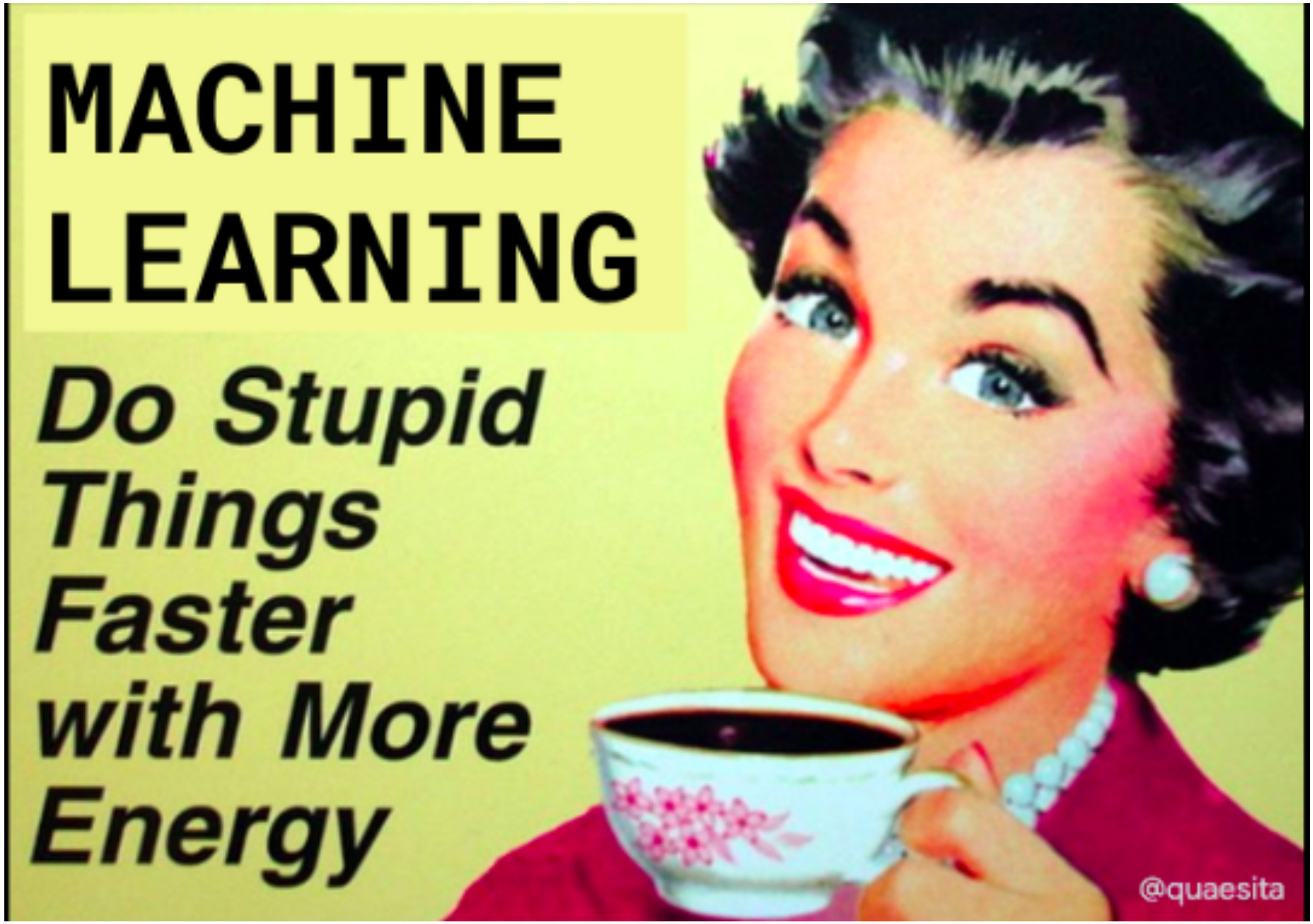
REF: https://towardsdatascience.com/battle-of-the-deep-learning-frameworks-part-i-cff0e3841750

MAS…...



REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

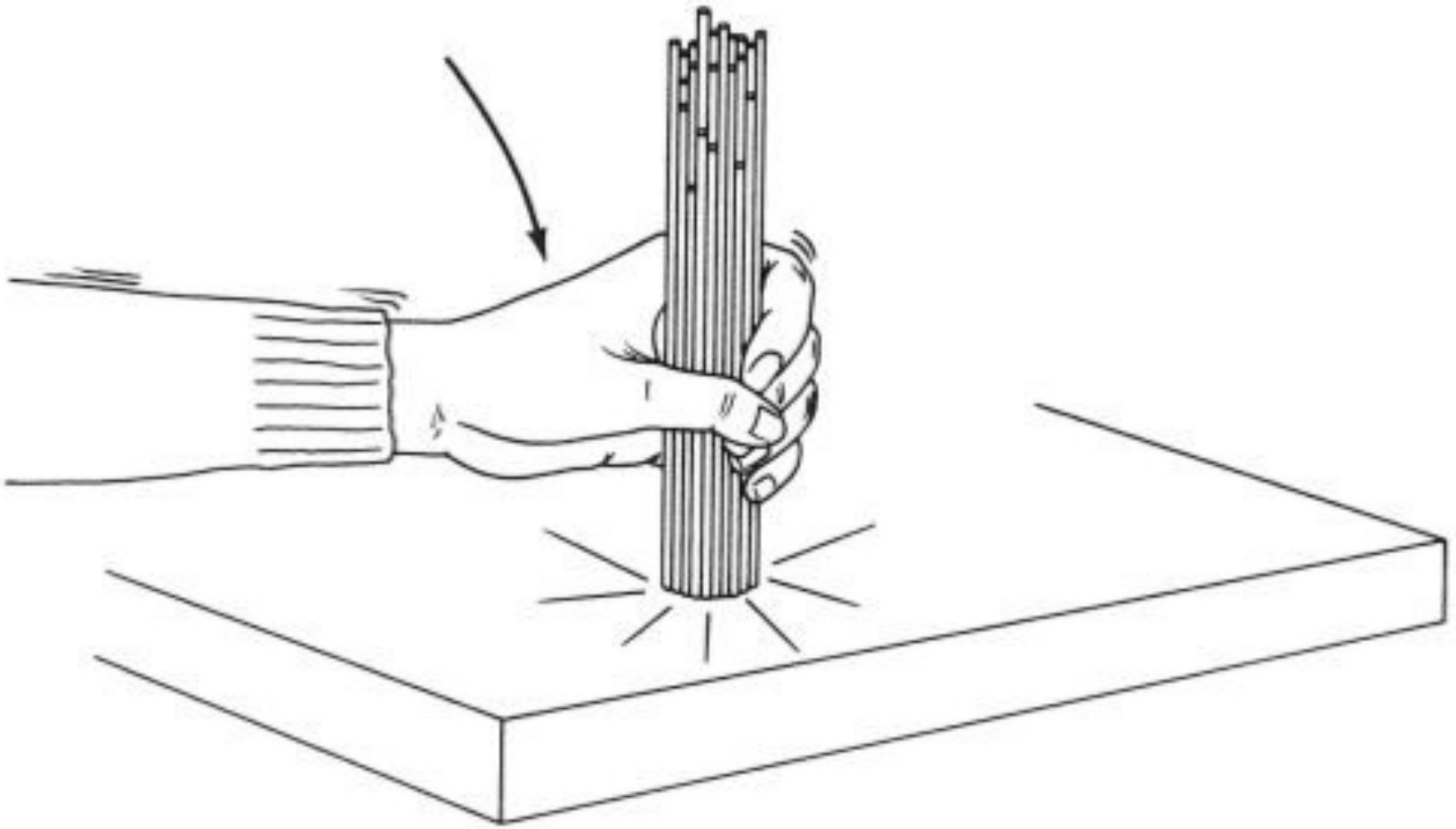
APRENDIZAGEM DE MÁQUINA



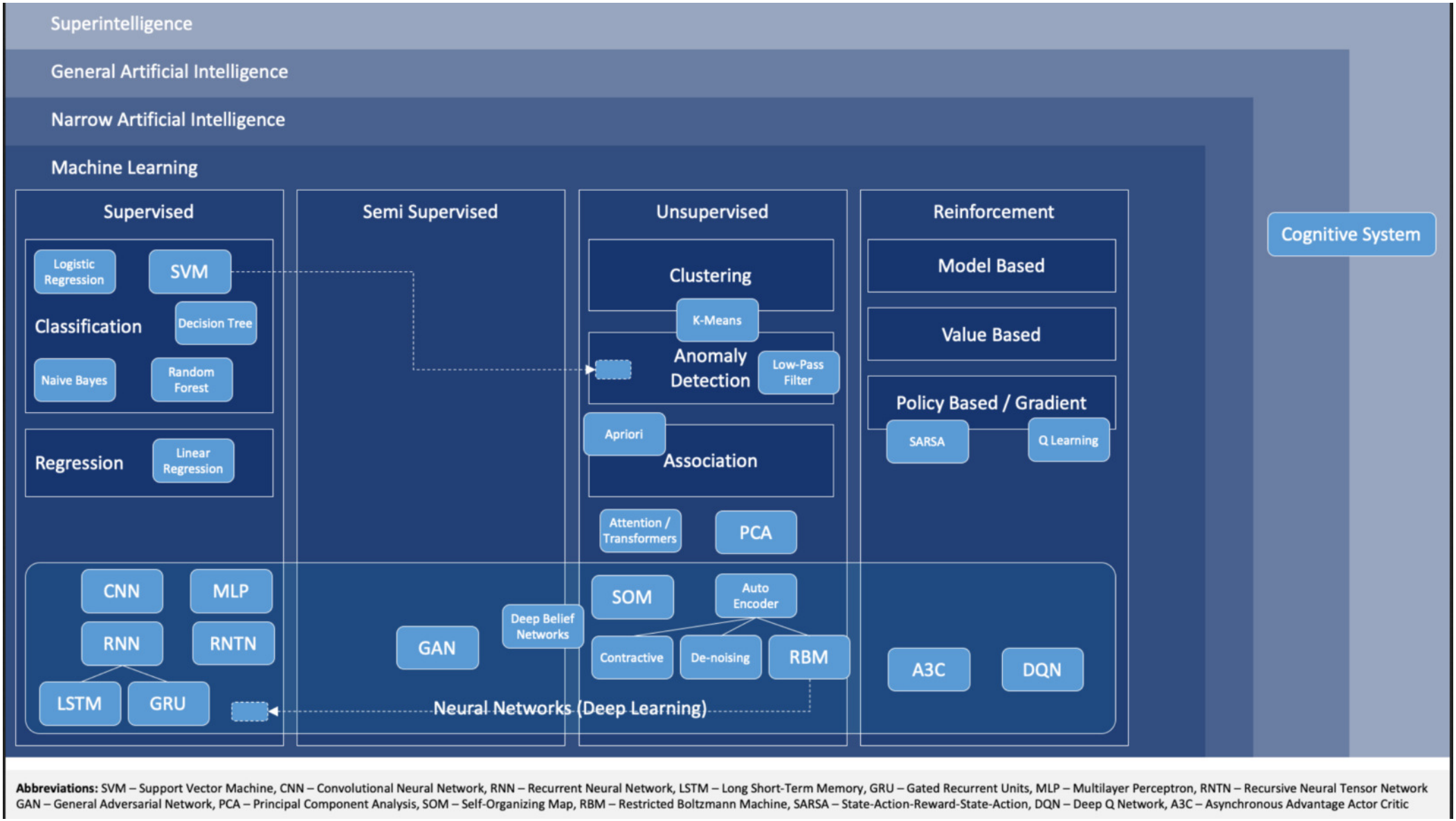
REF: https://towardsdatascience.com/artificial-intelligence-do-stupid-things-faster-with-more-energy-379aa6bac220

LIMITES

ORDENAÇÃO SPAGHETTI:
Computar dessa maneira requer CORPO FÍSICO
(como a Computação DNA)

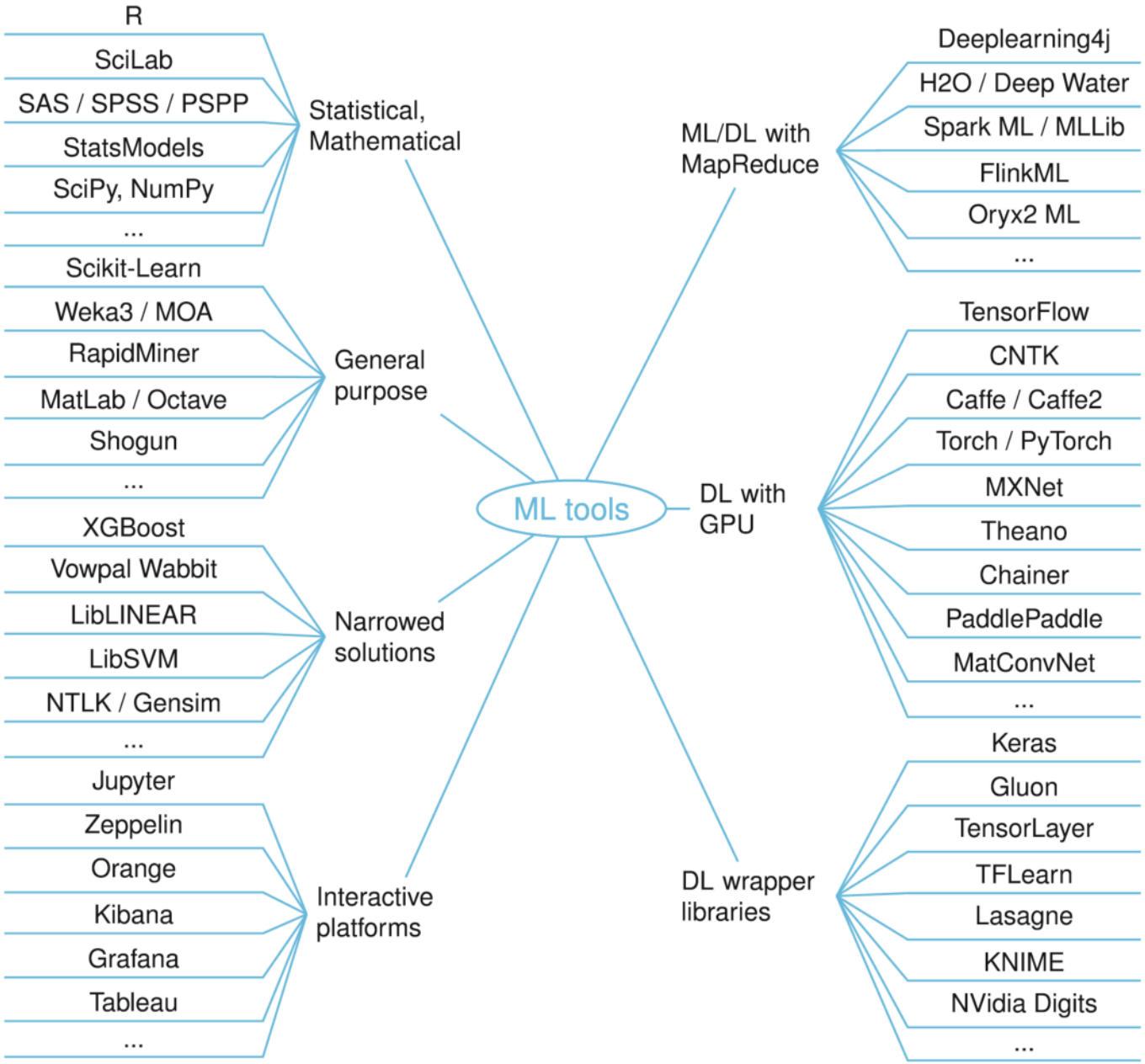


REF: https://towardsdatascience.com/the-nature-of-human-intelligence-from-a-computer-science-perspective-30f6e9c09146



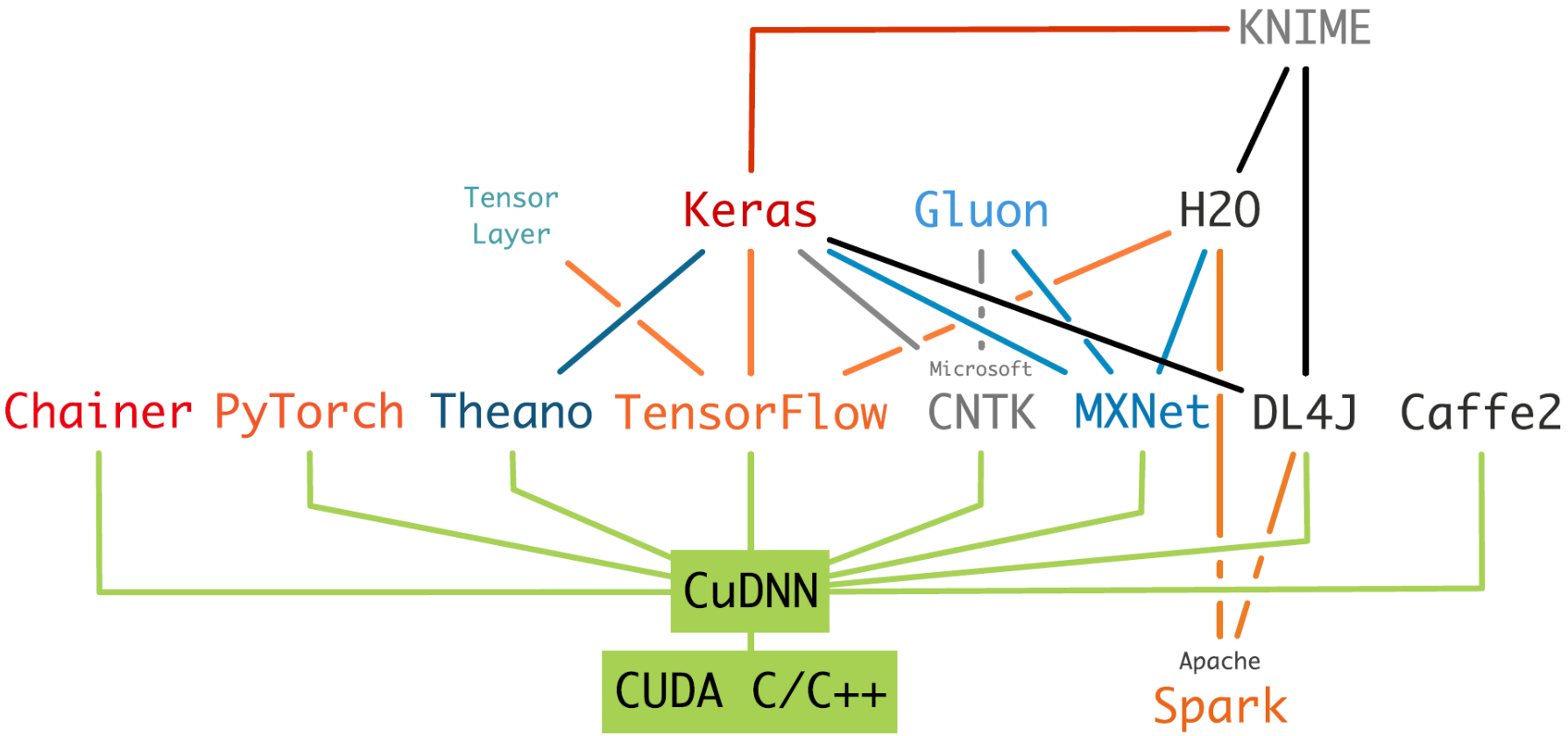
REF: https://towardsdatascience.com/artificial-intelligence- framework-a-visual-introduction-to-machine-learning-and-ai-d7e36b304f87 - Under Creative Commons CCBY-ND 4.0

ML-2019: Bibliotecas e Arcabouços (Frameworks)



REF:Artificial Intelligence Review (2019) 52:77–124
https://doi.org/10.1007/s10462-018-09679-

Deep Learning -2019: Bibliotecas e Arcabouços (Frameworks) em níveis de abstração



REF:Artificial Intelligence Review (2019) 52:77–124
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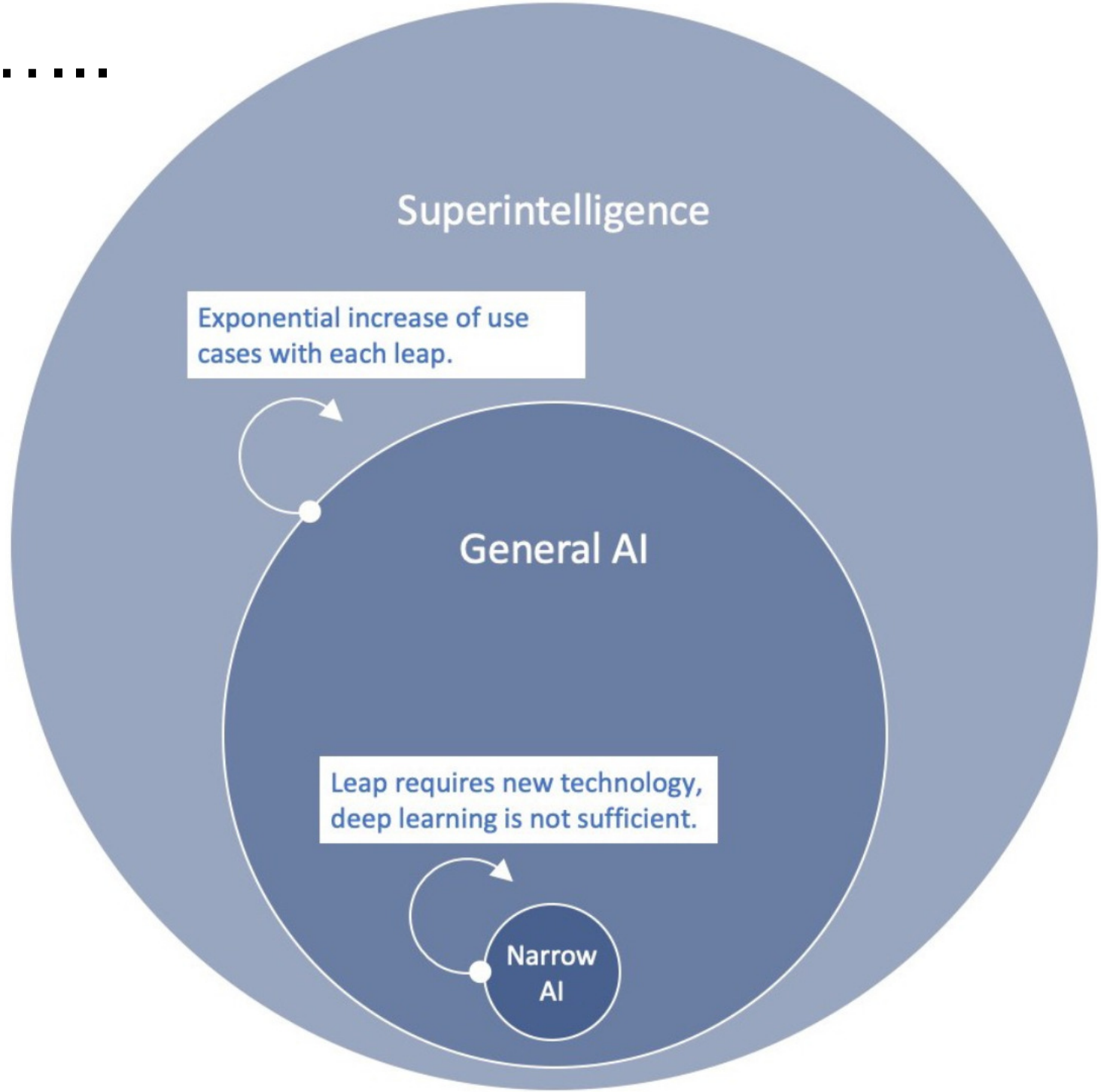
Deep Learning -2019: Bibliotecas e Arcabouços (Frameworks) com suporte GPU

Tool	Licence	Written in	Computation graph	Interface	Popularity	Usage	Creator (notes)
TensorFlow (Numerical framework)	Open source, Apache 2.0	C++, Python	Static with small support for dynamic graph	Python, C++[a], Java[a], Go[a]	Very High Growing very fast	Academic Industrial	– **Google**
Keras (Library)	Open source, MIT	Python	Static	Python Wrapper for TensorFlow, CNTK, DL4J, MXNet, Theano	High Growing very fast	Academic Industrial	F. Chollet
CNTK (Framework)	Open source, Microsoft permissive license	C++	Static	Python, C++, BrainScript, ONNX	Medium Growing fast	Academic Industrial Limited mobile solution	– **Microsoft**
Caffe (Framework)	Open source, BSD 2-clause	C++	Static	C++, Python, MatLab	High Growing fast	Academic Industrial	Y. Jia **BAIR**
Caffe2 (Framework)	Open source, Apache 2.0	C++	Static	C++, Python, ONNX	Medium-low Growing fast	Academic Industrial Mobile solution	Y. Jia **Facebook**
Torch (Framework)	Open source, BSD	C++, Lua	Static	C, C++, LuaJIT, Lua, OpenCL	Medium–low Growing low	Academic Industrial	R. Collobert, K. Kavukcuoglu, C. Farabet
PyTorch (Library)	Open source, BSD	Python, C	Dynamic	Python, ONNX	Medium Growing very fast	Academic Industrial	A. Paszke, S. Gross, S. Chintala, G. Chanan
MXNet (Framework)	Open source, Apache 2.0	C++	Dynamic dependency scheduler	C++, Python, Julia, MatLab, Go, R, Scala, Perl, ONNX	Medium Growing fast	Academic Industrial	– **Apache**
Chainer (Framework)	Open source, Owners permissive license	Python	Dynamic	Python	Low Growing low	Academic Industrial	– **Preferred Networks**
Theano (Numerical framework)	Open source, BSD	Python	Static	Python	Medium-low Growing low	Academic Industrial	Y. Bengio **University of Montreal**

[a]Not fully covered

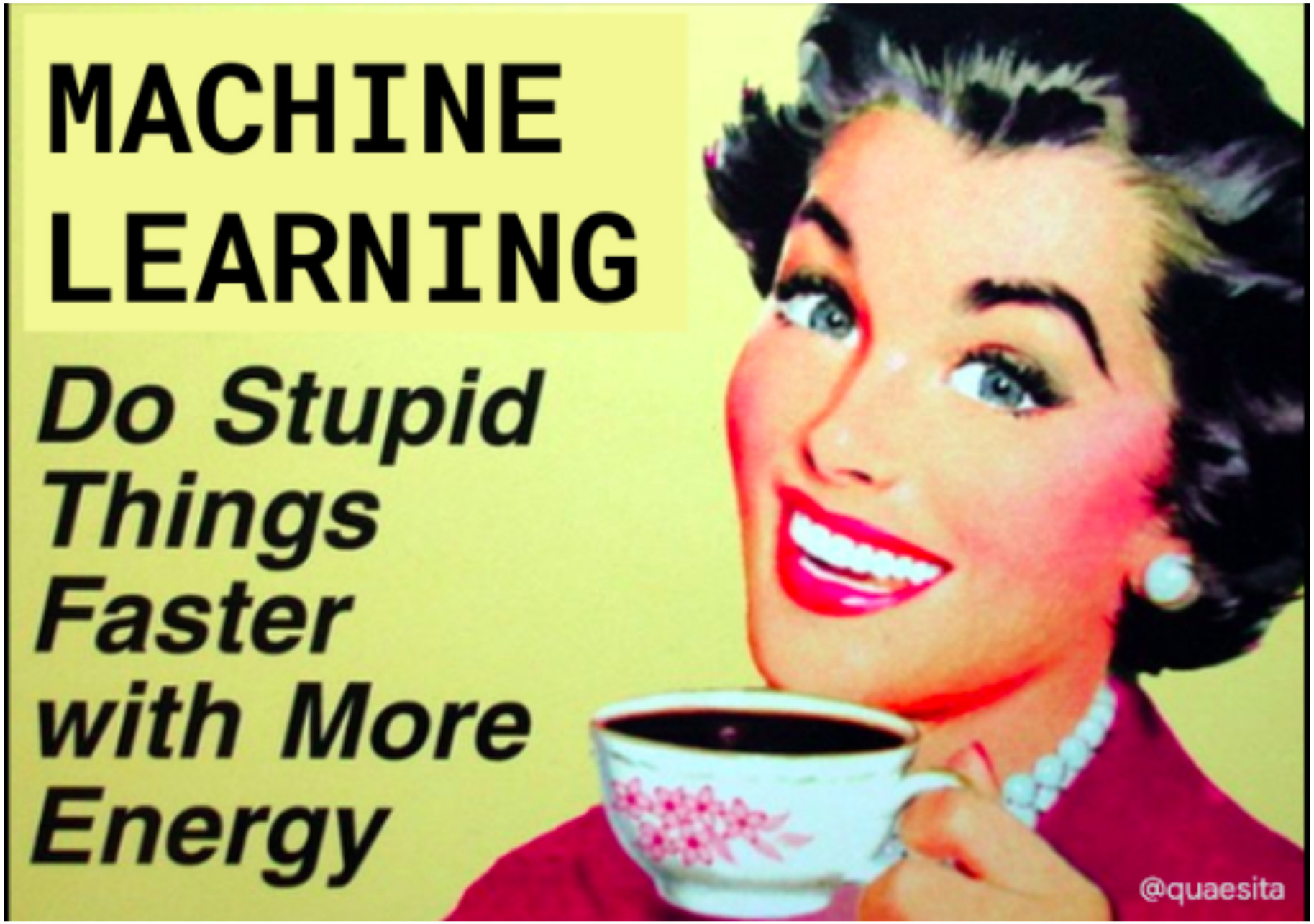
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MAS……



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APRENDIZAGEM DE MÁQUINA



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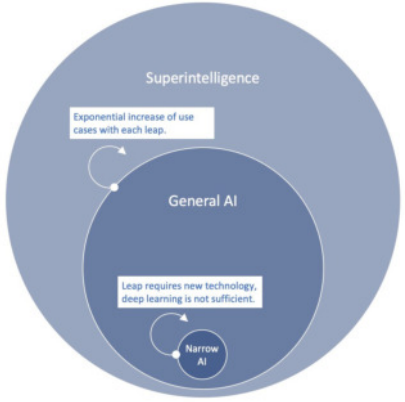


Precisamos ESTUDAR MAIS PROFUNDAMENTE
para sair da arena movediza das “buzz words”

REF: https://towardsdatascience.com/the-nature-of-human-intelligence-from-a-computer-science-perspective-30f6e9c09146



EXEMPLO: como funciona ?



COMO FUNCIONA EXATAMENTE ?!

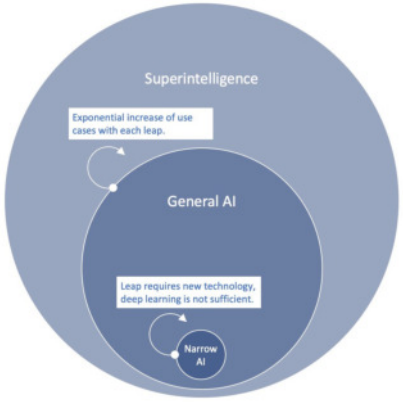
Superintelligence
General Artificial Intellige
Narrow Artificial Intelliger
Machine Learning
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Abbreviations: SVM – Support Vector Machine, CNN – Convolutional Neural Network, RNN – Recurrent Neural Network, LSTM – Long Short-Term Memory, GRU – Gated Recurrent Units, MLP – Multilayer Perceptron, RNTN – Recursive Neural Tensor Network GAN – General Adversarial Network, PCA – Principal Component Analysis, SOM – Self-Organizing Map, RBM – Restricted Boltzmann Machine, SARSA – State-Action-Reward-State-Action, DQN – Deep Q Network, A3C – Asynchronous Advantage Actor Critic

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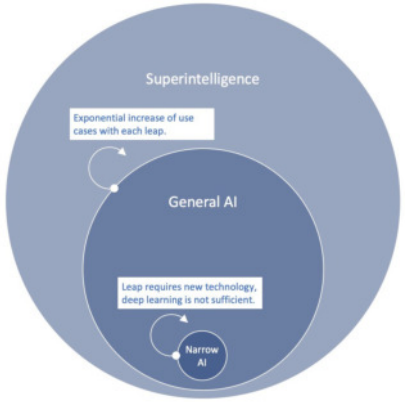
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EXEMPLO: como funciona ?



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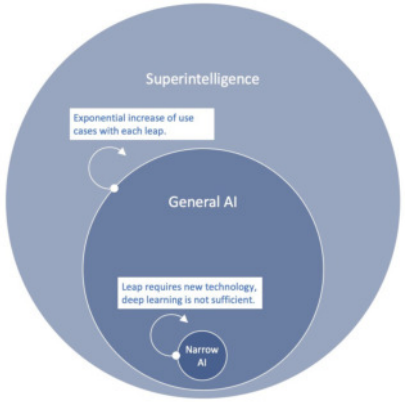
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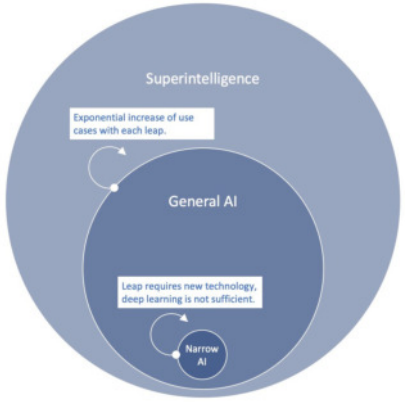
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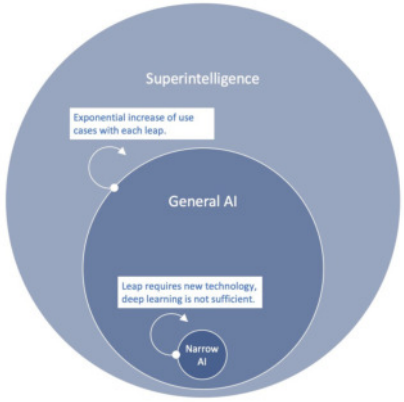
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DQN

Neural Networks (Deep Learning)

Abbreviations: SVM – Support Vector Machine, CNN – Convolutional Neural Network, RNN – Recurrent Neural Network, LSTM – Long Short-Term Memory, GRU – Gated Recurrent Units, MLP – Multilayer Perceptron, RNTN – Recursive Neural Tensor Network GAN – General Adversarial Network, PCA – Principal Component Analysis, SOM – Self-Organizing Map, RBM – Restricted Boltzmann Machine, SARSA – State-Action-Reward-State-Action, DQN – Deep Q Network, A3C – Asynchronous Advantage Actor Critic

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